

Foundations of 4D/6D Object Capture for Virtual Environments

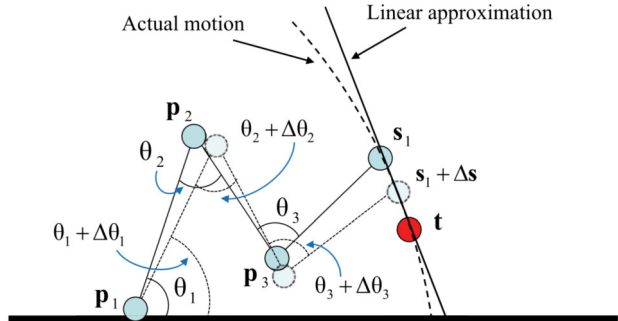
Motion Tasks and Outlook

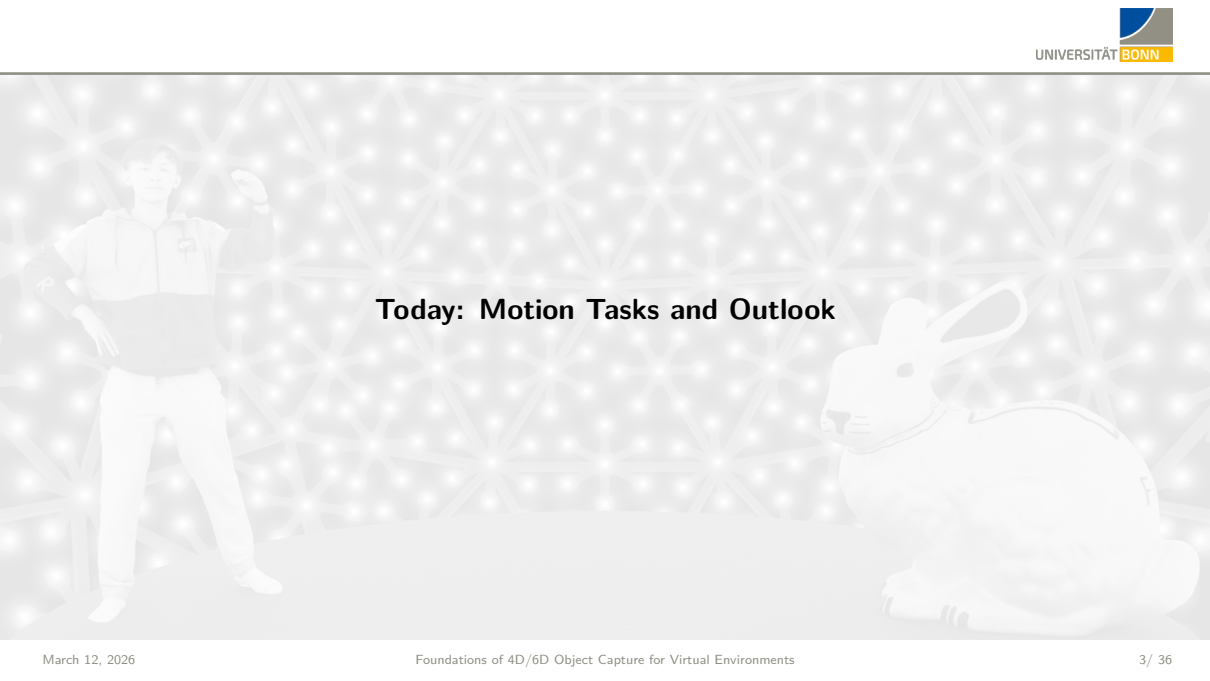


March 12, 2026



→ Inverse Kinematics



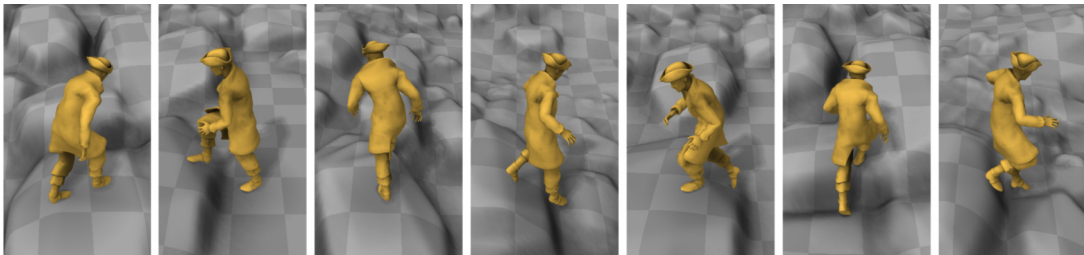


Today: Motion Tasks and Outlook

Problem statement:

Adjust (or create) a motion sequence according to the **environment**¹

- ▶ Move whole body when accelerating or decelerating
- ▶ Raise knees when going uphill
- ▶ Jump down from a higher spot to a lower one
- ▶ Use upper body for balancing across a thin beam

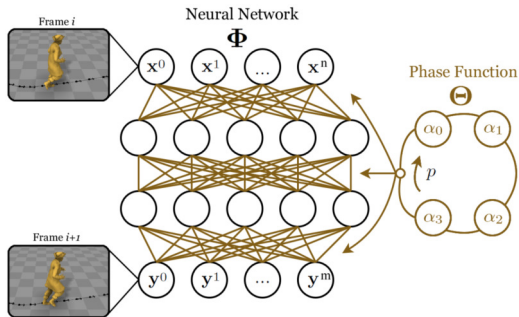


¹D. Holden et al. "Phase-Functioned Neural Networks for Character Control". In: *ACM Transactions on Graphics (TOG)* (2017).

Learning such motion transitions directly leads to stiff, unnatural results

Idea: Use periodic phase function to model timing of the motion cycle

- Change network weights based on current phase
- Each set of weights captures a (slightly) different learned function to adapt the motion



<https://youtu.be/Ul0Gilv5wvY>

Problem statement:

Transfer and adjust motion from one skeleton to another one with different properties²³

- ▶ Different length of limbs
- ▶ New joint restrictions
- ▶ Adapt to different surface geometry → Avoid penetration
- ▶ Adapt to environment (e.g. keep feet on the floor)

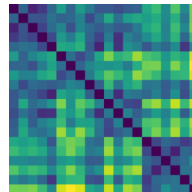


²R. Villegas et al. "Neural Kinematic Networks for Unsupervised Motion Retargeting". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2018.

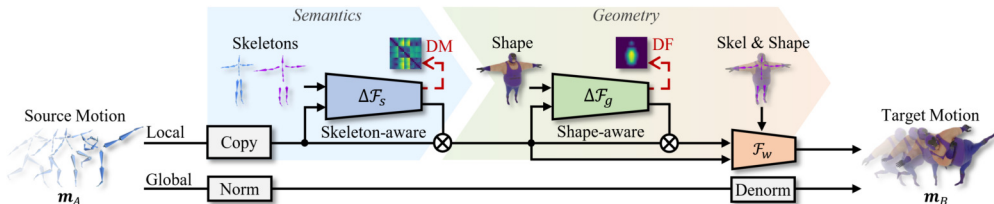
³J. Zhang et al. "Skinned Motion Retargeting with Residual Perception of Motion Semantics & Geometry". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023.

Idea:

- Copy motion and adjust it step by step
- Each step should follow original joint rotations, positions, constraints, ...
- (Semantics) Preserve relative joint distances by matching "normalized distance matrix"
- (Geometry) Repel interpenetrating surface mesh but also attract hands to keep potential contacts via "distance field"
- Learn a weighting function to balance semantics and geometry



Normalized distance matrix

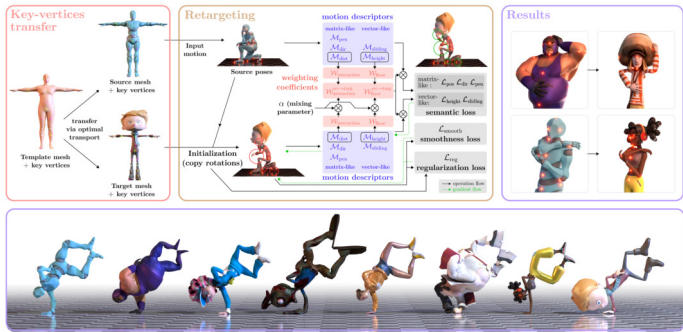


Adjust the motion (joint rotations and root position) only based on key vertices of the surface mesh⁴

→ Direct optimization of the motion without a neural network involved

Reduce the error of all interesting effects:

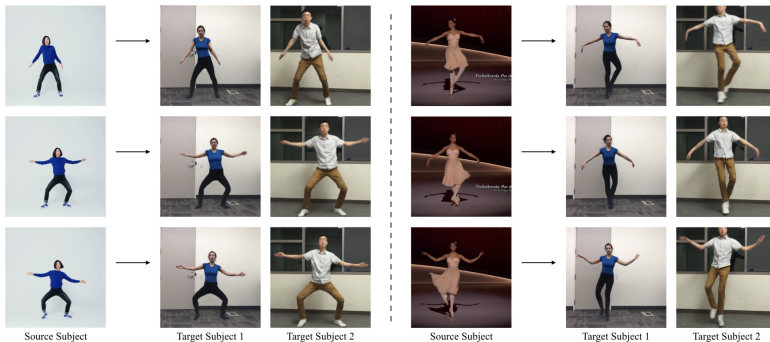
- ▶ Smooth motion
- ▶ Feet on ground without sliding
- ▶ No penetration
- ▶ ...



⁴T. Cheynel et al. "ReConForM: Real-time Contact-aware Motion Retargeting for more Diverse Character Morphologies". In: *Computer Graphics Forum (CGF)* (2025).

Retargeting is also possible from 2D data only⁵

→ Lower quality but no 3D skeleton required (use OpenPose to get 2D skeleton)



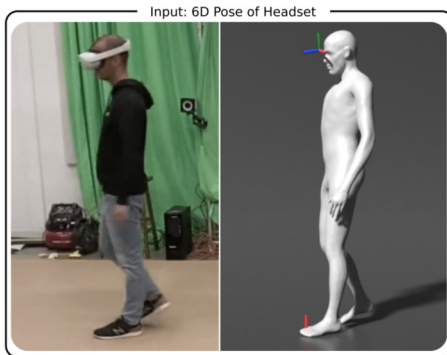
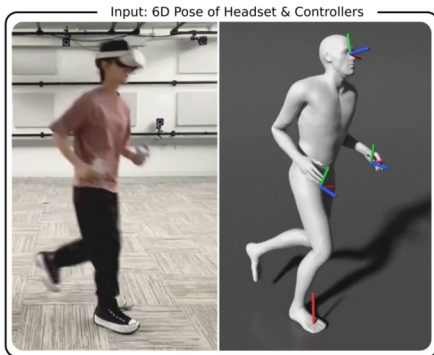
<https://youtu.be/PCBTZh41Ris>

⁵C. Chan et al. "Everybody Dance Now". In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2019.

Problem statement:

Estimate the full body pose / motion from **sparse** sensor data⁶

→ Usually just from sensors in a VR headset and 2 controllers



⁶A. Winkler et al. "QuestSim: Human Motion Tracking from Sparse Sensors with Simulated Avatars". In: *SIGGRAPH Asia Conference Papers*. 2022.

Idea: Learn an optimal "policy" describing how an agent acts in a given state

Given:

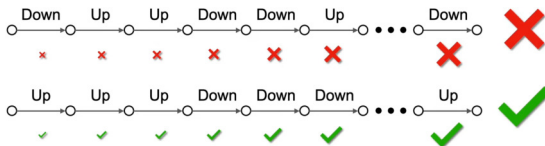
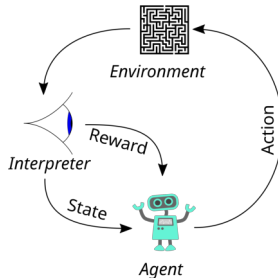
- Set of agent states $\mathcal{S} = \{\dots, s_i, \dots\}$ in an environment
- Set of possible actions $\mathcal{A} = \{\dots, a_i, \dots\}$

Objective: Learn policy

$$\pi(a_t | s_t) \in [0, 1] \quad (\text{probability of } a_t \text{ given } s_t)$$

that maximizes the **expected cumulative reward** over future states

$s_{t+1}, s_{t+2}, s_{t+3}, \dots$



How can we train such a policy with Reinforcement Learning?

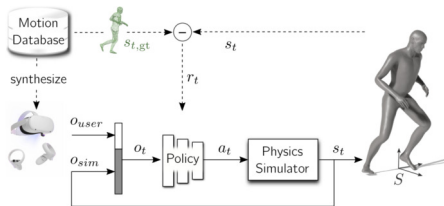
→ Generate sensor data from motion capture datasets where information of *all* joints available

Task:

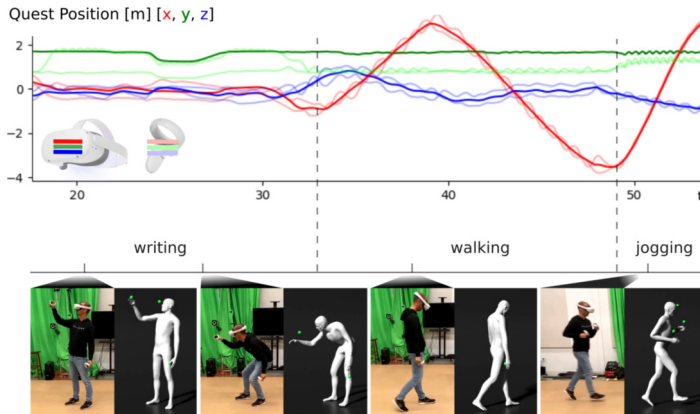
1. Sample pose $s_{t,\text{gt}}$ from database
2. Synthesize sparse headset observations o_{user} from $s_{t,\text{gt}}$
3. Predict action and run physics simulator to get state s_t
4. Compute reward

$$r_t = w_1 r(\mathbf{q}_t) + w_2 r(\dot{\mathbf{q}}_t) + w_3 r(\mathbf{x}_t) + w_4 r(\dot{\mathbf{x}}_t) + w_5 r_f$$

- ▶ Small difference in joint angles $\mathbf{q}$ and angular velocities $\dot{\mathbf{q}}$
- ▶ Small difference in joint positions $\mathbf{x}$ and linear velocities $\dot{\mathbf{x}}$
- ▶ Regularize foot motion to penalize abrupt decrease in vertical contact force $f_y \in \mathbb{R}$



Results: Common *patterns* in the motion sequences can be learned → e.g. oscillating motion in x , z (walking)



Here, y denotes vertical dimension shown in images

Results: Involved physics simulation allows for real-time interaction with virtual objects



<https://youtu.be/CkTHsz6Ldas>

Other architectures also possible with its own advantages and disadvantages

AvatarPoser⁷ combines a **Transformer** model with Inverse Kinematics for the arms

- Only indirect temporal consistency
- Self penetration and object penetration not explicitly handled

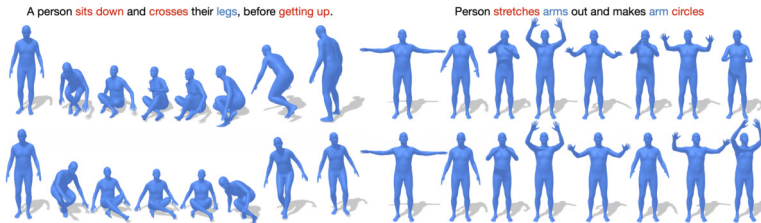


⁷J. Jiang et al. "AvatarPoser: Articulated Full-Body Pose Tracking from Sparse Motion Sensing". In: *European Conference on Computer Vision (ECCV)*. 2022.

Problem statement:

Create a desired motion sequence from scratch, e.g. just using a text prompt (text to motion)

- ▶ Train a **Variational Autoencoder (VAE)** for motion to get an Encoder-Decoder structure⁸
- ▶ Train a **matching encoder** for text inputs
- ▶ **Inference:** 1. Encode a text prompt, 2. Decode it with the motion decoder part



→ No guarantee to satisfy any constraint for plausibility

⁸C. Guo et al. "Generating Diverse and Natural 3D Human Motions from Text". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022.

Idea: Train a network to remove the noise from an image (or other input models)⁹

Training Stage

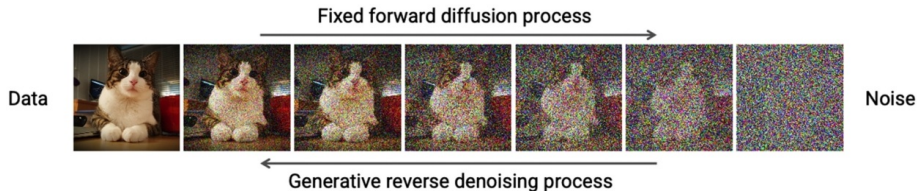
- ▶ Start with latent (encoded) vector x_0 of input
- ▶ Successively add Gaussian noise $z_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

$$x_{t+1} = \sqrt{1 - \alpha_t} x_t + \sqrt{\alpha_t} z_t$$

- ▶ Learn to predict the total accumulated noise that has been applied to the input

Inference Stage

- ▶ Start with random noise $x \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- ▶ Evaluated network and "subtract" that noise
→ Moves x_t towards averaged/blurred version
- ▶ Add Gaussian noise $z_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to x_{t-1}
→ Corresponds to sampling the full distribution rather than collapsing to the mean



⁹J. Ho et al. "Denoising Diffusion Probabilistic Models". In: *Advances in Neural Information Processing Systems (NeurIPS)* 33 (2020).

Note: Diffusion models can be conditioned on additional parameters

Text to motion via diffusion

- ▶ Train a denoiser to iteratively refine motion parameters¹⁰
- ▶ Condition denoising step with a text prompt
- ▶ Result depends on initial noise → Generate varying motion naturally

Example:

A person kicks with their left leg

<https://youtu.be/rVkIDj5wgjs?t=63>

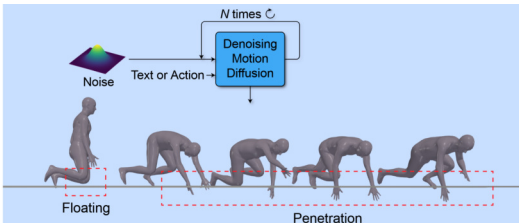
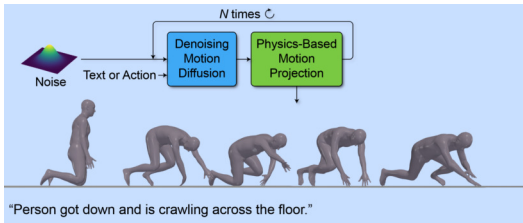


¹⁰G. Tevet et al. "Human Motion Diffusion Model". In: *International Conference on Learning Representations (ICLR)*. 2023.

Note: Diffusion models can be conditioned on additional parameters

Text to motion via diffusion + physics simulation

- Include a physics-based motion projection step after denoising¹¹
- Try to mimic the input motion but for a physically simulated character



<https://nvlabs.github.io/PhysDiff/>

¹¹Y. Yuan et al. "PhysDiff: Physics-Guided Human Motion Diffusion Model". In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2023.

Observation: All of these methods want to fulfill the same constraints

Depending on the approach, different constraints are achievable and one usually has to choose:

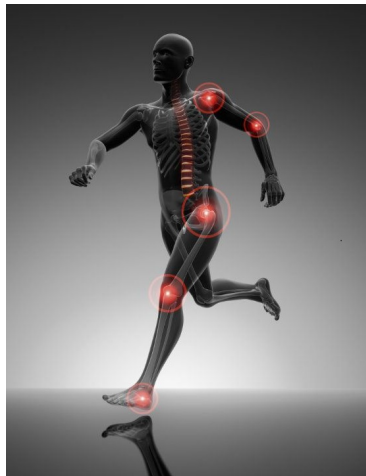
- ▶ Smooth motion
- ▶ Respect joint constraints
- ▶ Feet on ground
- ▶ No foot sliding
- ▶ No self-penetration
- ▶ No scene/object penetration (e.g. with the floor)
- ▶ Physically realistic (balance, realistic momentum)
- ▶ Adapt to environment (e.g. uneven floor or obstacles)

Summary:

- ▶ No method is already perfect
- ▶ Motion transfer to 2D characters
- ▶ Hybrid methods (e.g. for film industry)
- ▶ (Medical) motion analysis
 - ▶ Monitor illness
 - ▶ Analyze sports performance
 - ▶ Track general health and fitness

→ More information/details in the lecture
MA-INF 2202 Computer Animation

Thanks to **Prof. Dr. Björn Krüger** (bkrueger@uni-bonn.de)
for providing information from his lecture



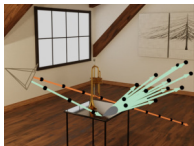
Outlook

Throughout this course, we covered several topics:

- ▶ **Week 1:** Linear Transformations → Cameras → Scene Representation → Reconstruction Methods
- ▶ **Week 2:** Human Models → Colors and Rendering → Differentiable Rendering → Deep Learning
- ▶ **Week 3:** Neural Radiance Fields → Gaussian Splatting → Inverse Kinematics → Motion Tasks

But there is more to come beyond this course:

Our Research



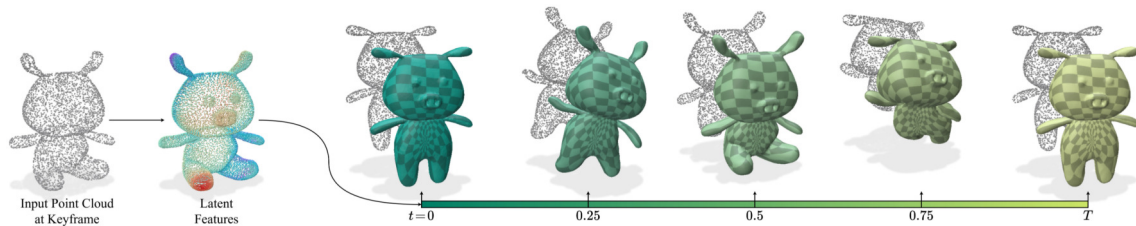
Our Projects



Related Modules



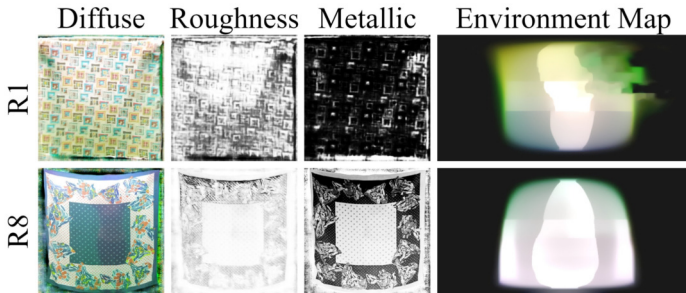
Goal: Learn spatially *smooth* and temporally *coherent* deformations from dynamic point clouds¹²



<https://cg.cs.uni-bonn.de/publication/kaltheuner-2026-neu-pig>

¹²J. Kaltheuner, H. Dröge, et al. "Neu-PiG: Neural Preconditioned Grids for Fast Dynamic Surface Reconstruction on Long Sequences". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.

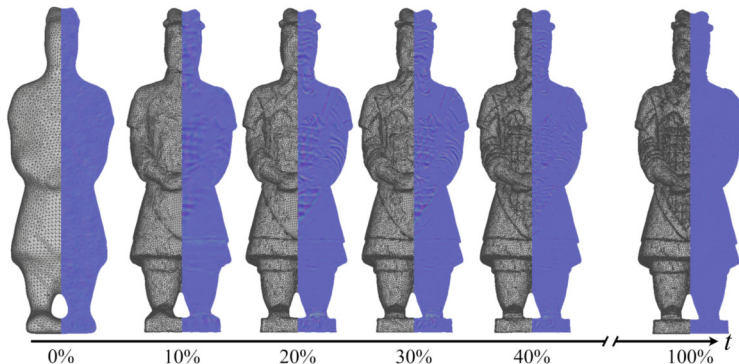
Goal: Estimate *physical and material parameters* of cloth from monocular video¹³



<https://cg.cs.uni-bonn.de/publication/stotko-2025-saft>

¹³D. Stotko and R. Klein. "SAFT: Shape and Appearance of Fabrics from Template via Differentiable Physical Simulations from Monocular Video". In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025.

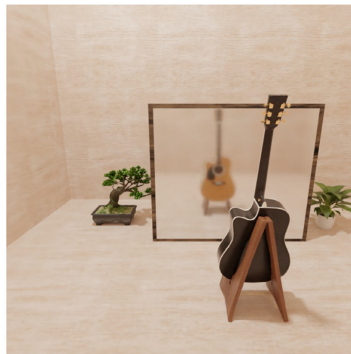
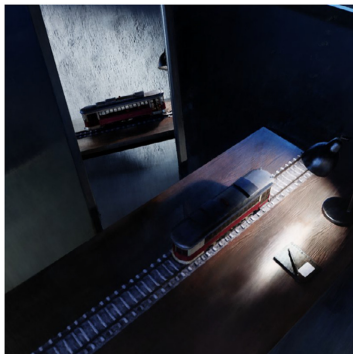
Goal: Reconstruct *highly detailed but still compact* mesh geometry with BRDF texture¹⁴



<https://cg.cs.uni-bonn.de/publication/kaltheuner-2025-rosa>

¹⁴J. Kaltheuner, P. Stotko, et al. "ROSA: Reconstructing Object Shape and Appearance Textures by Adaptive Detail Transfer". In: *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2025.

Goal: Model mirrors in NeRF based on *sparse user annotations*¹⁵ → **from Master's Thesis**



<https://cg.cs.uni-bonn.de/publication/holland-2024-tram-nerf>

¹⁵L. V. Holland, R. Bliersbach, et al. "TraM-NeRF: Tracing Mirror and Near-Perfect Specular Reflections Through Neural Radiance Fields". In: *Computer Graphics Forum* (2024).

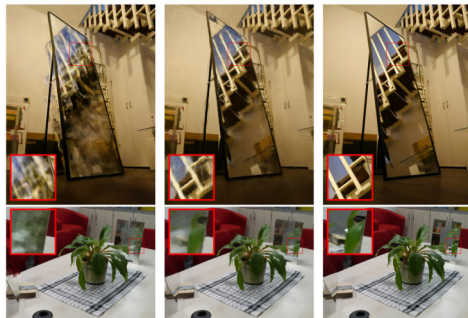
Goal: Detect and handle mirrors in NeRF *without additional (user) input*¹⁶



Best Baseline
MS-NeRF [112]

Ours

Ground Truth



Best Baseline
Mip-NeRF 360 [3]

Ours

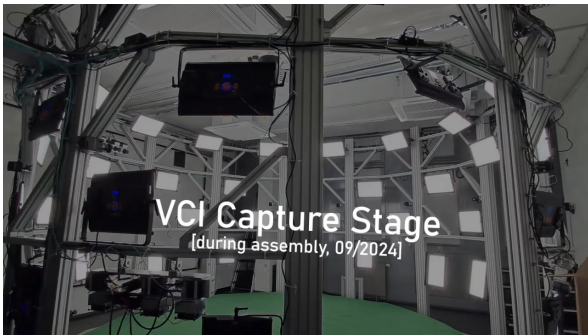
Ground Truth

<https://cg.cs.uni-bonn.de/publication/holland-2025-nerfs-are-mirror-detectors>

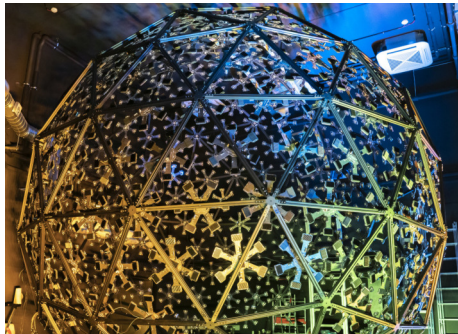
¹⁶L. V. Holland, M. Weinmann, et al. "NeRFs are Mirror Detectors: Using Structural Similarity for Multi-View Mirror Scene Reconstruction with 3D Surface Primitives". In: *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2025.

Goal: Build *technology platform* to push forward developments in Computer Graphics, Virtual Reality or Human-Computer Interaction → For Creatives, Researchers, and Entrepreneurs

Capture Stage 1



Capture Stage 2



<https://cg.cs.uni-bonn.de/project/visual-computing-incubator>

Goal: With VCI data, conduct *interdisciplinary research* of human behavior in virtual reality environments



<https://cg.cs.uni-bonn.de/project/invirtuo-4-0>

Goal: With VCI data, build tools to generate *emotional, animatable avatars* from text prompts, sketches, etc.



<https://cg.cs.uni-bonn.de/project/gen-aivatar>

Summer Semester (SS)

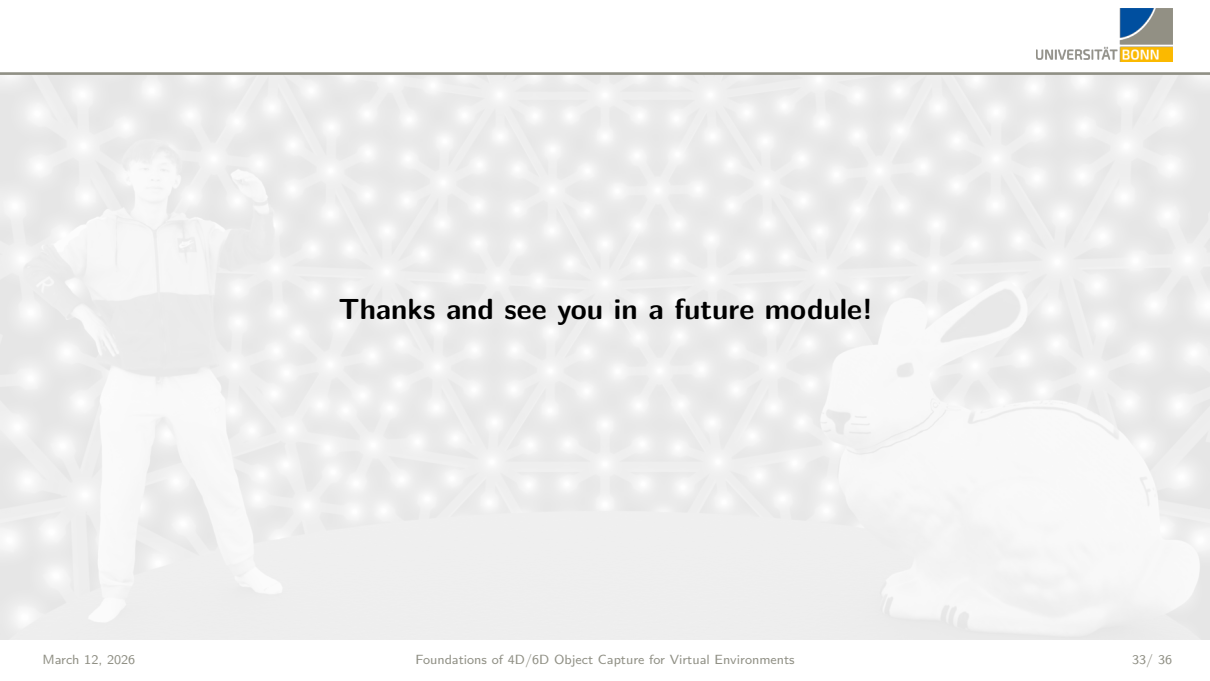
- ▶ MA-INF 2209 [Advanced Topics in Computer Graphics I](#)
→ Path Tracing, Importance Sampling
- ▶ MA-INF 2214 [Computational Photography](#)
→ Optics, Light Fields
- ▶ MA-INF 2202 [Computer Animation](#)
→ Motion Synthesis, Facial Animations

Winter Semester (WS)

- ▶ MA-INF 2310 [Advanced Topics in Computer Graphics II](#)
→ Geometric Mesh Processing, Shape Spaces

Every Semester (SS + WS)

- ▶ MA-INF 2207 [Seminar](#)
 - ▶ MA-INF 2308 [Lab](#)
 - ▶ MA-INF 0401 [Master's Thesis](#)
- Wide range of topics in Visual Computing



Thanks and see you in a future module!

- [1] C. Chan, S. Ginosar, T. Zhou, and A. A. Efros. “Everybody Dance Now”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2019.
- [2] T. Cheynel, T. Rossi, B. Bellot-Gurlet, D. Rohmer, and M.-P. Cani. “ReConForM: Real-time Contact-aware Motion Retargeting for more Diverse Character Morphologies”. In: *Computer Graphics Forum (CGF)* (2025).
- [3] C. Guo, S. Zou, X. Zuo, S. Wang, W. Ji, X. Li, and L. Cheng. “Generating Diverse and Natural 3D Human Motions from Text”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022.
- [4] J. Ho, A. Jain, and P. Abbeel. “Denoising Diffusion Probabilistic Models”. In: *Advances in Neural Information Processing Systems (NeurIPS)* 33 (2020).
- [5] D. Holden, T. Komura, and J. Saito. “Phase-Functioned Neural Networks for Character Control”. In: *ACM Transactions on Graphics (TOG)* (2017).
- [6] L. V. Holland, R. Bliersbach, J. U. Müller, P. Stotko, and R. Klein. “TraM-NeRF: Tracing Mirror and Near-Perfect Specular Reflections Through Neural Radiance Fields”. In: *Computer Graphics Forum* (2024).

- [7] L. V. Holland, M. Weinmann, J. U. Müller, P. Stotko, and R. Klein. “NeRFs are Mirror Detectors: Using Structural Similarity for Multi-View Mirror Scene Reconstruction with 3D Surface Primitives”. In: *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2025.
- [8] J. Jiang, P. Streli, H. Qiu, A. Fender, L. Laich, P. Snape, and C. Holz. “AvatarPoser: Articulated Full-Body Pose Tracking from Sparse Motion Sensing”. In: *European Conference on Computer Vision (ECCV)*. 2022.
- [9] J. Kaltheuner, H. Dröge, M. Plack, P. Stotko, and R. Klein. “Neu-PiG: Neural Preconditioned Grids for Fast Dynamic Surface Reconstruction on Long Sequences”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.
- [10] J. Kaltheuner, P. Stotko, and R. Klein. “ROSA: Reconstructing Object Shape and Appearance Textures by Adaptive Detail Transfer”. In: *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2025.
- [11] D. Stotko and R. Klein. “SAFT: Shape and Appearance of Fabrics from Template via Differentiable Physical Simulations from Monocular Video”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025.

- [12] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-Or, and A. H. Bermano. "Human Motion Diffusion Model". In: *International Conference on Learning Representations (ICLR)*. 2023.
- [13] R. Villegas, J. Yang, D. Ceylan, and H. Lee. "Neural Kinematic Networks for Unsupervised Motion Retargeting". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2018.
- [14] A. Winkler, J. Won, and Y. Ye. "QuestSim: Human Motion Tracking from Sparse Sensors with Simulated Avatars". In: *SIGGRAPH Asia Conference Papers*. 2022.
- [15] Y. Yuan, J. Song, U. Iqbal, A. Vahdat, and J. Kautz. "PhysDiff: Physics-Guided Human Motion Diffusion Model". In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2023.
- [16] J. Zhang, J. Weng, D. Kang, F. Zhao, S. Huang, X. Zhe, L. Bao, Y. Shan, J. Wang, and Z. Tu. "Skinned Motion Retargeting with Residual Perception of Motion Semantics & Geometry". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023.